

Teleoperated driving via private 5G for remote operation of robots in hazardous manufacturing environments, reducing on-site personnel exposure.

Manufacturing × Remote and autonomous operations × Private 5G / LTE · OS077 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
67 is the world moving	88 can we play, can we win	HIGH 0 named in the evidence	€2.9bn partial evidence · per year	LATER research and standards activi...	L0 14 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity enables manufacturers to remotely operate robots in hazardous environments via a private 5G network, reducing on-site personnel exposure. It targets manufacturing operations with high-risk areas such as chemical plants, foundries, or explosive atmospheres. The need is urgent due to recent demonstrations and research validating the technical feasibility of teleoperated driving over private 5G.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€4.0bn €504m – €26.0bn	€2.9bn €362m – €18.6bn	€171m €21.7m – €1.1bn	partial
Observed: tendered contract value, annualised	€20.9m €20.9m – €56.6m	€20.9m €20.9m – €56.6m	€1.3m €1.3m – €3.4m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting Private 5G / LTE	30.0 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_iotn2 · E_IOT1	2021
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	6.0 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Internet of Things by NACE Rev. 2 activity series E_IOT1 is used as a proxy for Private 5G / LTE: No private-network series; industrial IoT use as proxy. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

Observed: tendered contract value, annualised

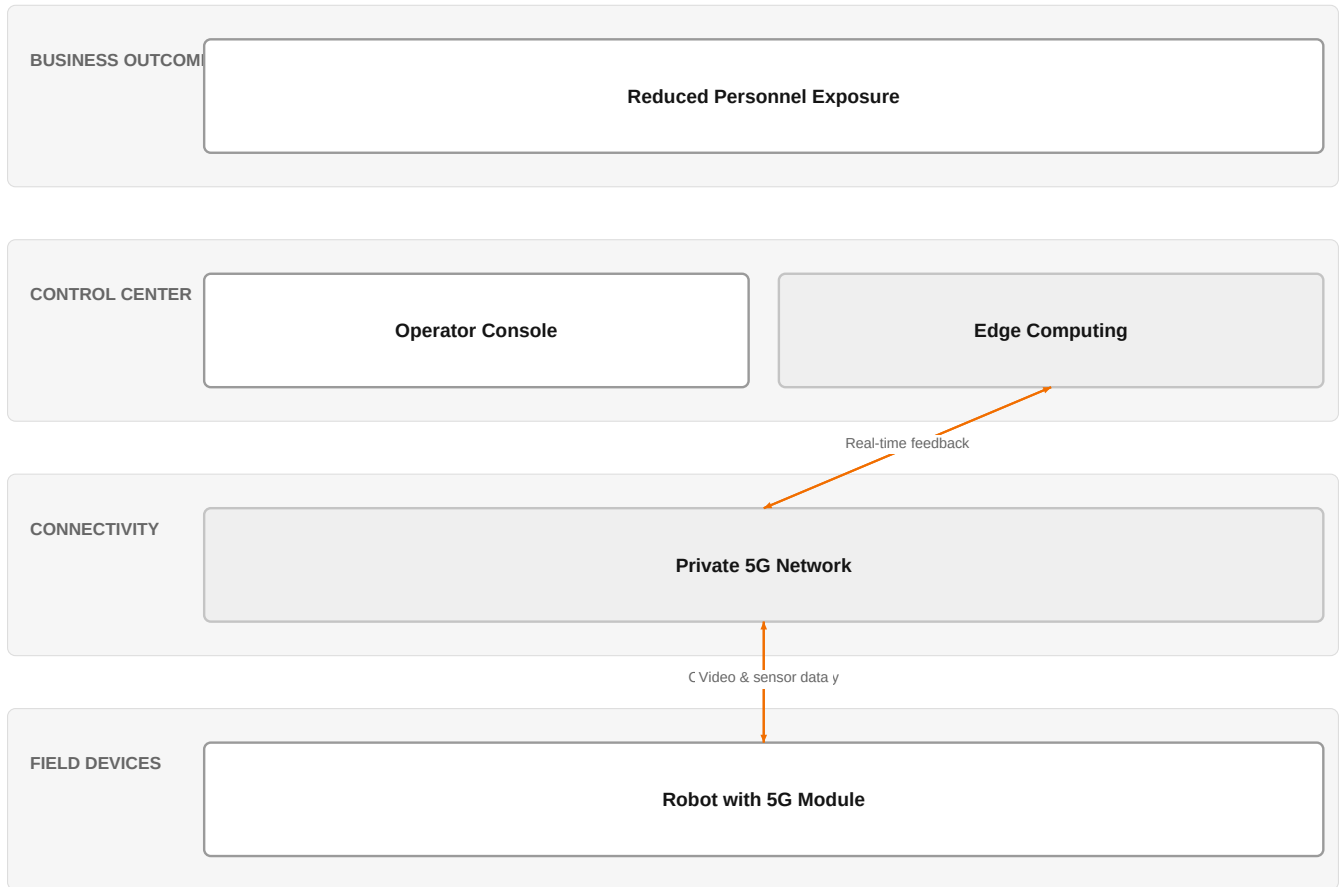
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	6.0 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a

market size.

The solution, in one picture

Teleoperated Driving on Private 5G



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how a private 5G network connects an operator console to a robot in a hazardous area, enabling safe remote operation with low latency.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Recent research demonstrates teleoperated driving via private 5G networks with remotely-driven robots, proving the concept's viability. Advances in cross-layer latency optimization for URLLC services in 5G MEC networks support latency-critical applications like industrial automation, addressing the strict latency requirements of remote operation. Additionally, deployment and coverage optimization methods for 5G-A industrial private networks address the challenges of dynamic terminals like mobile robots and AGVs in manufacturing environments.

Sources: SIG-B11748E56A0F 2026 IEEE 12th Interna 2026-06-29; SIG-18012D9BCACF Springer Science and B 2026-06-19; SIG-67F7CBD007F7 Applied Sciences 2026-05-22

Who buys, and why now

The primary buyer is the plant operations director or head of manufacturing, who is responsible for worker safety and operational continuity. The pain is the risk of injury or fatality in hazardous zones, which drives the need to reduce personnel exposure. The trigger is often a recent safety incident, an insurance premium increase, or new regulatory pressure to minimize on-site risks. The operations team feels the pain daily as they balance safety protocols with productivity targets.

Why it is hot now — every claim bound to a dated source

- Teleoperated driving via private 5G demonstrates remote operation of industrial robots. [SIG-B11748E56A0F]
- Demonstration of teleoperated driving via a private 5G network with remotely-driven robots. [SIG-B11748E56A0F]

- Cross-layer latency optimization for URLLC services in 5G MEC networks using adaptive network slicing supports latency-critical applications like industrial automation. [SIG-18012D9BCACF]
- Teleoperated driving via private 5G networks has been demonstrated, enabling remote operation of robots. [SIG-B11748E56A0F]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would deliver a private mobile network (L0) tailored for teleoperated driving, leveraging its position as a Leader in the Gartner Magic Quadrant for 4G and 5G Private Mobile Network Services. The engagement would start with a site assessment and network design, followed by deployment of the private 5G/LTE infrastructure, and then integration with the customer's robots and control systems. Orange's IoT and Digital experts would support the implementation, and the solution would be certified to ISO 27001 and TISAX standards where applicable.

Why Orange can win

Orange's leadership in the Gartner Magic Quadrant for private mobile networks provides a strong credibility anchor. The availability of IoT, Digital, and Defense and security experts allows for a comprehensive solution that addresses both connectivity and security. Additionally, Orange's certifications (ISO 27001, TISAX) are critical for manufacturing customers with strict compliance requirements. However, the assets are not unique; competitors also offer private 5G, so differentiation must come from the depth of expertise and the ability to integrate with existing systems.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Private mobile networks	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
Samsung	partner	L2	partner provides the required technology	unconfirmed
Huawei	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
Gartner Magic Quadrant 2026 — 4G and 5G Private Mobile Network Services (Leader)	analyst position	SUP	analyst recognition covers this technology or use case	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical, different use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

HIGH competitive intensity (score 8.0; 0 of 8 listed players appear in this topic's own evidence)

Crowded, with heavyweight incumbents already visible in the evidence. Win on a specific differentiator or do not bid.

Competitor	Category	Position here	Basis	Evidence
BT Business	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—

Deutsche Telekom / T-Systems	Telecom operator peer	sells Private 5G / LTE; active in Manufacturing; competes in Connectivity Solutions	structural	—
NTT DATA	Global systems integrator	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Proximus	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Swisscom	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Telefonica Tech	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Telia Company	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—
Verizon Business	Telecom operator peer	sells Private 5G / LTE; competes in Connectivity Solutions	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What specific hazardous areas in your facility currently require human presence, and what tasks are performed there?
- 2 Have you experienced any safety incidents or near-misses in those areas in the past 12 months?
- 3 What is your current approach to reducing personnel exposure in hazardous zones (e.g., automation, remote monitoring)?
- 4 Do you already have any robots or automated guided vehicles (AGVs) in your facility that could be retrofitted for remote operation?
- 5 What are your latency and reliability requirements for remote control of equipment?
- 6 Who in your organization would be the champion for a project like this, and what budget would they have access to?

Objections you will meet

They will say	You can answer
We already have Wi-Fi and it works fine for our current needs.	Wi-Fi can work for some applications, but teleoperated driving requires deterministic low latency and high reliability that Wi-Fi cannot guarantee in dynamic industrial environments. Private 5G provides the URLLC capabilities needed for safe remote operation.
This is too new and unproven for our critical operations.	It's true that teleoperated driving over private 5G is an emerging use case, but the technology has been demonstrated in research and is backed by industry standards. We can start with a pilot in a less critical area to prove the value before scaling.
The cost of deploying a private 5G network is too high.	The initial investment is significant, but it can be offset by reduced insurance premiums, fewer accidents, and increased productivity. We can also explore a managed service model to spread the cost over time.

Risks and unknowns

The main risk is that teleoperated driving in hazardous environments is still in early stages; the evidence is primarily from research demonstrations, not large-scale commercial deployments. The customer may be hesitant to adopt a technology that is not yet proven in their specific environment. Additionally, the integration with existing robots and control systems is complex and may require significant customization. Unknowns include the actual latency performance in real-world manufacturing settings and the total cost of ownership compared to traditional automation.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a hazardous manufacturing bid, referencing the Colgate-Palmolive, Mondelez International, and Saint-Gobain Glass references, and highlighting ISO 27001 and TISAX certifications, along with Samsung and Huawei partnerships, to differentiate our private mobile networks offer.

Sales	In a meeting with a manufacturing plant operations director, say: 'We have demonstrated teleoperated driving via private 5G networks, enabling remote operation of robots to reduce personnel exposure in hazardous environments. Our private mobile networks, backed by our Leader position in the Gartner Magic Quadrant, can help you achieve this.'
Strategist / Innovator	Commission a deep-dive brief on teleoperated driving via private 5G for hazardous manufacturing, leveraging Orange Group researchers and IoT experts to validate the cross-layer latency optimization approach and assess the maturity of the technology for later time horizon.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-B11748E56A0F	2026-06-29	2026 IEEE 12th Internatio...	1	Teleoperated Driving via a Private 5G Network: Demonstration with Remotely-Driven Robots
SIG-18012D9BCACF	2026-06-19	Springer Science and Busi...	1	Cross-Layer Latency Optimization for URLLC Services in 5G MEC Networks Using Adaptive Network S...
SIG-67F7CBD007F7	2026-05-22	Applied Sciences	1	Deployment and Coverage Optimization Methods for Base Stations Under Multi-Type Terminal Scenar...
SIG-9798339F3F2D	2026-01-01	IEEE Access	1	Multi-Connectivity for Industrial Automation: Achieving Deterministic Reliability Through Priva...
SIG-BC7FE33C6859	2025-11-28	openalex.org	1	Research on the Implementation of New Communication Technologies to Improve Quality and Stabili...
SIG-3056D9FEB764	2025-11-17	openalex.org	1	Data-Driven, Real-Time Diagnostics of 5G and Wi-Fi Networks Using Mobile Robotics
SIG-873E0DC9CA64	2025-11-07	International Journal of ...	1	Architecting and Analyzing Private 5G/LTE Networks for Ultra-Reliable and Deterministic Industr...
SIG-5E2C87F5E70E	2025-10-14	openalex.org	1	5G and UWB Integration for Robot Collaboration
SIG-D031822F4DA6	2025-09-09	openalex.org	1	Closed-Loop Network Control for Industrial Edge Computing: A Telemetry-Driven and AI-Based Appr...
SIG-D9E7FFE6A4A5	2025-08-21	openalex.org	1	Head Smart - Revolutionizing Industrial Safety with IoT
SIG-02E085CB6CF2	2026-07-03	Telecom Review Africa	2	Private 5G and Industrial IoT: From Connectivity to Competitive Advantage - Telecom Review Afri...
SIG-CDD2DA807ACB	2026-08-11	arxiv.org	3	A Neural Network Based Teleoperation for Remote Controlled Vehicles
SIG-77F434506B29	2026-07-29	framerusercontent.c om	3	Industry Brief: Private 5G for Manufacturing and Industrial Sites [pdf]
SIG-AF0B9D624A57	2026-06-30	arxiv.org	3	Latency-Sensitive 5G RAN Slicing for Deterministic Aperiodic Traffic in Smart Manufacturing

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:56:42+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.